

# Basic Tools Constructing Insurable Risk Portfolios

Colombia Workshop

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## Outline of Presentation

- Risk Measures
- Risk Retention Functions
- Dependent Risks
- Constrained Optimization

## Recall the Problem

- There are  $p$  risks,  $X_j, j = 1, \dots, p$
- For each risk, the risk owner
  - retains some risk,  $g_j(X_j)$
  - and cedes risk to another party  $X_j - g_j(X_j)$
- The cost of ceding these risks is denoted as  $RTC$  (risk transfer cost)
  - For example,  $RTC = E \left\{ \sum_{j=1}^p X_j - g_j(X_j) \right\}$
- The total retained risk is  $g(\mathbf{X}) = \sum_{j=1}^p g_j(X_j)$ 
  - Capture this uncertainty with a risk measure,  $RM[g(\mathbf{X})]$
- Our goal is to quantify this trade-off for different types of retention plans

# Risk Measures

## 2.1 Measures of Uncertainty

Let  $X$  denote a generic risk.

- As benchmarks, we use classic measures of uncertainty
  - the variance,  $\text{Var}(X) = \text{E}[X - \text{E}(X)]^2$
  - the standard deviation (positive square root of the variance)
- The value at risk is  $\text{VaR}_\alpha[X] = \inf\{x : F_X(x) \geq \alpha\}$ 
  - We also denote this as  $\text{VaR} = F_\alpha^{-1}$ , the  $\alpha$  quantile
  - To distinguish the value of risk ( $\text{VaR}$ ) from the variance ( $\text{Var}$ ), we follow a common practice in the literature and capitalize the **R** in the value at **R**isk.
- The expected shortfall,  $\text{ES}_\alpha(X) = \text{E}(X|X > F_\alpha^{-1})$ 
  - the expected amount given that the loss exceeds the  $\text{VaR}$ .

## Quantiles and Value at Risk

- Random variable  $X$ , distribution function  $F$ , inverse  $F^{-1}$ .
  - For a fraction  $0 < q < 1$ , we can write the  $q$ th quantile as  $q_F = F^{-1}(q)$ .
- To simulate outcomes from  $F$ , use the **inverse transform method**.
  - Start by simulating a uniform random number,  $U_r$ . The convention is the range is  $[0, 1]$ .
  - Next, define  $X_r = F^{-1}(U_r)$ .
  - Repeat  $r = 1, \dots, R$  for a large number  $R$  replications.
- This method works well for *dependent uniform variables*, as will be seen.
- For additional background on quantiles, see for example, [Loss Data Analytics, Section 4.1.1.3](#).
- For additional background on simulating random variables, see for example, [Loss Data Analytics, Section 6.1](#)

## Quantile-based Risk Measures

- $VaR$  and  $ES$  are types of quantile-based risk measures
- For comparison purposes, the book also employs two other quantile-based risk measures
  - The *GlueVaR*, defined as

$$GlueVaR = \omega_1 VaR_\alpha + \omega_2 ES_\alpha + \omega_3 ES_{\alpha+\beta}.$$

- The **range value at risk** is defined as

$$RVaR_{(\alpha,\beta)}(X) = \begin{cases} \frac{1}{\beta} \int_{\alpha}^{\alpha+\beta} VaR_a(X) da & \text{if } \beta > 0 \\ VaR_\alpha(X) & \text{if } \beta = 0 \end{cases}.$$

# Risk Retention Functions

- Think about  $p$  risks (such as 15), denoted as  $X_1, \dots, X_p$
- Each risk is subject to an insurance contract that specifies:
  - $g_j(X_j; \theta)$  - the amount retained
  - $X_j - g_j(X_j; \theta)$  - the amount transferred to another party

These contracts can be complex, so we start at the foundations:

- Suppose that  $g(x; \theta = c) = c \cdot x$ , a *coinsurance* or *quota-share*
  - For example, your insurer pays 80% of your prescription drugs expenses, you pay 20%
- Suppose that  $g(x; \theta = u) = \min(x, u)$ , known as *deductible* or *excess of loss*
  - For example, you retain claims from your auto accident up to 250, the insurer pays the excess
- I refer to  $c, u, \theta$  as **risk retention** parameters

### 2.2.1 A Risk Retention Function

An important special case that covers many situations of interest is

$$g(x; d, c, u) = \begin{cases} 0 & \text{if } x < d \\ c(x - d) & \text{if } d \leq x < u \\ c(u - d) & \text{if } x \geq u \end{cases} .$$

Here,  $d$  is the deductible,  $c$  is the coinsurance amount, and  $u$  is the upper limit of coverage. This function can also be expressed as

$$g(x; d, c, u) = c(x \wedge u - x \wedge d) .$$

The wedge symbol  $\wedge$  is used for the minimum operator and is defined as  $x \wedge z = \min(x, z)$ .

### 2.2.3 Risk Measures

Quantiles preserve location and scale transformations. For example,

$$\text{VaR}_\alpha(a_1 X - a_0) = a_1 \times \text{VaR}_\alpha(X) - a_0, \quad \text{for } a_1 > 0.$$

This holds for all quantile-based measures, including *ES*, *RVaR*, and *GlueVaR*.

For an upper limit only policy ( $d = 0, c = 1$ ), one can check that

$$F_{g(X;u)}(z) = \Pr(\min(X, u) \leq z) = \begin{cases} F(z) & \text{if } z < u \\ 1 & \text{if } z \geq u \end{cases}.$$

So

$$\text{VaR}_\alpha[g(X; u)] = \begin{cases} F_\alpha^{-1} & \text{if } \alpha < F(u-) \\ u & \text{if } \alpha \geq F(u-) \end{cases}.$$

Using the figure for the distribution of  $F_{g(X;\theta)}(z)$ , check that

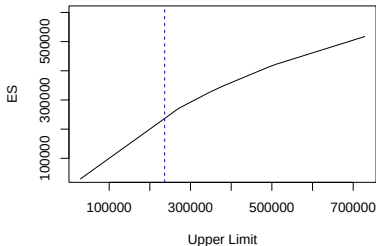
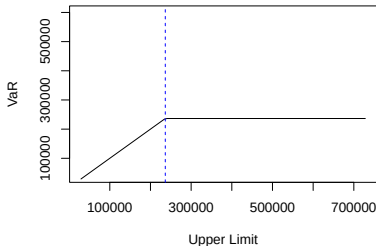
$$VaR_{\alpha}[g(X)] = \begin{cases} 0 & \text{if } \alpha < F(d) \\ c(F_{\alpha}^{-1} - d) & \text{if } F(d) \leq \alpha < F(u-) \\ c(u - d) & \text{if } \alpha \geq F(u-) \end{cases} .$$

In the same way, check that

$$ES_{\alpha}[g(X)] = \begin{cases} \frac{c}{1-\alpha} \{E(X \wedge u) - E(X \wedge d)\} & \text{if } \alpha < F(d) \\ \frac{c}{1-\alpha} \{E(X \wedge u) - E(X \wedge F_{\alpha}^{-1}) \\ \quad + (1-\alpha)(F_{\alpha}^{-1} - d)\} & \text{if } F(d) < \alpha < F(u) \\ c(u - d) & \text{if } F(u) < \alpha \end{cases} .$$

## Example 2.1. Property Fund - Continued.

- Assume no deductible ( $d = 0$ ) and no coinsurance ( $c = 1$ ).
- A confidence level of  $\alpha = 0.99$ .
- $\widehat{VaR}_{0.99} = 236,427$ , marked with blue dashed vertical lines
- The left-hand panel of this figure shows a change in the slope for the  $VaR$  whereas the right-hand panel shows that the  $ES$  is smoother at this point.



## General Risk Retention Functions

- Through simulation strategies, we will have the ability to analyze complex risk transfer agreements.
  - Recall that  $g(\mathbf{X}; \theta)$  represents a risk owner's retained risk after a risk transfer agreement.
- In a one-day workshop, we will focus on basic functions such as quota share and excess of loss.
- A summary of general risk retention functions is Section 7.5 of the book.

We will go through a quick run through of a few of these now.

## Separate Contracts for each Risk

- Could postulate a different contract for each risk.
  - The amount retained for the  $j$ th risk is  $g_j(X_j; \theta_j)$ .
  - With an additive structure among retained risks

$$g(\mathbf{X}; \theta) = \sum_{j=1}^p g_j(X_j; \theta_j).$$

- Both quota share and excess of loss are special cases.
  - Could also have some risks follow quota share, others excess of loss.
  - Could use  $\theta = (d, c, u)$  where  $d$  is the deductible,  $c$  is the coinsurance amount, and  $u$  is the upper limit of retention.
- Working with additive structures will simplify the numerical convergence of our optimization procedures.

## Combining Risks

### Combined Deductibles from Bundled Insurance Contracts

- In personal insurance, combined deductibles are common when insurers “bundle” together several types of insurance.
- For example, in travel insurance  $X_1$  might represent the loss due to luggage damage and  $X_2$  is the loss due to personal injury.
- Then, a type of insurance policy covers

$$Y_{insurer} = [(X_1 - d_1)_+ + (X_2 - d_2)_+ - d_{combined}]_+.$$

- $d_1$  is the deductible for luggage claims and  $(X_1 - d_1)_+$  is the luggage excess.
- $d_2$  is the deductible for personal injury and  $(X_2 - d_2)_+$  is the personal injury excess.
- The insurer is responsible for the excess over a combined deductible,  $d_{combined}$ , over these two separate excesses.

### Example. Global Protections Reinsurance Coverage.

- Consider an agreement where the reinsurer's portion of the total risk is

$$Y_{reinsurer} = \min \left( \sum_{j=1}^p \min[(X_j - u_{1,j})_+, u_{2,j}], AAL \right).$$

- $u_{1,j}$  represents a reinsurer's deductible,
- $u_{2,j}$  represents is upper limit for the  $j$ th risk, and
- $AAL$  stands for the annual aggregate limit  $AAL$ .

### Example. Separate Limits by Peril.

- A cause of loss is known as *peril*
- It is common for there to exist special arrangements such as sublimits made for each peril.
- For example, one might introduce a sublimit  $u_1$  that limits how much of risk  $X_1$  can be transferred.
  - Then, if  $X_2$  represents all other risks and  $u_{overall}$  is an overall limit, we can use

$$Y_{firm} = (X_1 \wedge u_1 + X_2) \wedge u_{overall}$$

to express the amount retained by a firm.

## Frequency Severity Approaches to Modeling Risks

- For risks that occur frequently, it is convenient to define risk transfers on a *per event* or *per occurrence of a claim* basis.
- the *collective risk* model - decompose the overall type  $j$  loss amount into
  - a number of claims in a year, say  $N_j$ , and
  - examine each individual claim outcome,  $X_{j1}, \dots, X_{jN_j}$ .
  - we express aggregate type  $j$  losses as  $X_j = \sum_{i=1}^{N_j} X_{ji}$ .
- Especially for high frequency risk types such as auto insurance, it is common to define risk transfer mechanisms (such as deductibles and upper limits) for each claim.
  - For example, a firm might retain claims up to  $u_j$  and so the total retained claims is

$$Y_{j,retained} = \sum_{i=1}^{N_j} X_{ji} \wedge u_j.$$

## Frequency Severity Approaches to Modeling Risks 2

- For some coverages such as property insurance, there may be many claims due to a common event such as earthquakes or hurricanes.
  - The amount transferred may be of the form

$$Y_{j,transfer} = \left[ \sum_{i \in \text{event}} (X_{ji} - d_j)_+ \right] \wedge u_j.$$

- For each claim,  $(X_{ji} - d_j)_+$  represents the amount in excess of a deductible  $d_j$ .
- The sum within the square braces is the total excess for all claims but the transferred amount is limited  $u_j$ .

# Dependent Risks

- In the insurance sector, risks are almost never independent.
- There are many ways to model dependence, latent variables being a common approach
- Instead, we will utilize **copulas**.
  - Still, much of the material in this course works well if you want to use another approach for modeling dependence.
- **Definition:** A *copula* is a multivariate distribution function with uniform marginals.
- Copulas enable us to work with multiple risks having different distributions,
  - perhaps the first one being gamma distributed, the second with a Pareto distribution, the third having a normal distribution.
  - Even though each marginal distribution may differ, we are able to link (or *couple*) them using a copula.

## 4.2.1 Copulas Defined

- Specifically, let  $\{U_1, \dots, U_p\}$  be  $p$  uniform random variables on  $(0, 1)$ . (**Not** necessarily independent!)
  - Their distribution function is called a **copula**

$$C(u_1, \dots, u_p) = \Pr(U_1 \leq u_1, \dots, U_p \leq u_p).$$

- To extend this, consider arbitrary *marginal* distribution functions  $F_1(\cdot), \dots, F_p(\cdot)$ .
  - Define a multivariate distribution function  $F$  using the copula such that

$$F(y_1, \dots, y_p) = C(F_1(y_1), \dots, F_p(y_p)).$$

- For any  $F(\cdot)$ , there *always* exists a copula (Sklar's Theorem).

## Two Examples of Copulas

*Frank's copula*, for  $p = 2$  variables

$$C_F(u_1, u_2) = \frac{1}{\gamma} \log \left( 1 + \frac{(\exp[\gamma u_1] - 1)(\exp(\gamma u_2) - 1)}{\exp(\gamma) - 1} \right).$$

- This is a bivariate distribution function.
  - Check that  $C(u_1, 1) = u_1 \implies$  marginals have a uniform (on  $[0, 1]$ ) distribution  $\implies C$  is a copula.
  - $\gamma$  is a dependence parameter.
  - Simple, but only useful for  $p=2$  variables.

- The *normal*, or *Gaussian*, copula. In  $p = 2$  dimensions, this is

$$C_N(u_1, u_2) = \Phi_2 \left( \Phi^{-1}(u_1), \Phi^{-1}(u_2) \right).$$

- The symbol  $\Phi(\cdot)$  represents the cumulative standard normal *df* and  $\Phi^{-1}(\cdot)$  is the corresponding quantile function
  - In R these can be evaluated using the functions `pnorm` and `qnorm`, respectively.
  - The symbol  $\Phi_2(\cdot)$  represents the a bivariate normal *df* that, in R, can be evaluated using the function `mvtnorm`.
  - The normal is also readily available in multiple dimensions.
- We use the normal copula extensively. But, for almost every application, other copulas readily apply.
  - The **extent** of the dependence matters, more so than the shape of the copula.

## 4.2.2 Copulas, Risk Retention, and Simulation

Copulas are useful for risk retention problems, especially when combined with simulation techniques.

### **Example 4.2.1. Excess of Loss with Two Risks**

- Let  $X_1$  be gamma and  $X_2$  be Pareto distributed.
- Use a normal copula with dependence parameter  $\rho_S = 0.30$ .
- The risk owner retains  $S(u_1, u_2) = X_1 \wedge u_1 + X_2 \wedge u_2$ .
- The distribution and moments are hard to compute analytically, easy with simulation

## Simulation Approach

- Using simulation, create a random realization of the bivariate uniform random variates  $(U_{1r}, U_{2r})$ . In the R package `copula`, we do this with the function `rCopula`.
- Use the inverse of the distribution function to create random realizations of the random variables,  $X_{1r} = F_1^{-1}(U_{1r})$  and  $X_{2r} = F_2^{-1}(U_{2r})$ .
  - For  $F_1^{-1}$ , in R we use the function `qgamma`.
  - For  $F_2^{-1}$ , in R we use the function `qpareto`.
- Create a random realization of the limited sum,  
 $S_r(u_1, u_2) = X_{1r} \wedge u_1 + X_{2r} \wedge u_2$ .
- Repeat this for a large number,  $R$ , of replications.

## Example 4.6 Illustrative Code

```
# Packages
library("actuar") ; library("copula")
# Set Simulation Parameters
nsim <- 20000
set.seed(202020)
# Marginal distribution parameters
risk1shape <- 1 ; risk1scale <- 10
risk2shape <- 1 ; risk2scale <- 10
# Normal, or Gaussian, Copula
norm.cop <- copula::normalCopula(param=0.3, dim = 2,'dispstr' ="un")
UCop <- copula::rCopula(nsim, norm.cop)
# Marginal Distributions
X1 <- qgamma(p=UCop[,1],shape = risk1shape, scale = risk1scale)
X2 <- actuar::qpareto(p=(UCop[,2]),shape = risk2shape,
                      scale = risk2scale)

# Risk Retention Function
g0 <- function(u1,u2){
  S <- pmin(X1,u1) + pmin(X2,u2)
  return( S )
}
```

# Constrained Optimization

## Algorithmic Framework - Risk Retention

- Define the retained risk

$$g(\mathbf{X}; \theta) = \sum_{j=1}^p g_j(X_j; \theta)$$

- Denote the uncertainty as  $RM[g(\mathbf{X}; \theta)]$ .
  - For the  $RM$  (risk measure), variance is a possibility. I typically use expected shortfall.
- The risk transferred is  $X_1 + \dots + X_p - g(\mathbf{X}; \theta)$ .
  - The cost of the transfer is denoted as  $RTC(\theta)$
  - It is evaluated using an insurance pricing principle, starting with an expectation (augmented for administrative costs)

## Risk Retention Optimization Problem

We seek to find those risk retention parameters in the vector  $\theta$  that

|                   |                              |                               |
|-------------------|------------------------------|-------------------------------|
| minimize $\theta$ | $RM[g(\mathbf{X}; \theta)]$  | <i>uncertainty</i>            |
| subject to        | $RTC(\theta) \leq RTC_{max}$ | <i>budget restriction</i>     |
|                   | $\theta_j \geq 0$            | <i>parameter restrictions</i> |

- Minimize the uncertainty in retained risks summarized by  $RM[g(\mathbf{X}; \theta)]$
- Limited to the aptly termed “budget” constraint  $RTC(\theta) \leq RTC_{max}$ .
  - The values of  $\theta$  that satisfy this constraint are known as the *feasible set*.
  - Could easily add other parameter restrictions (e.g.,  $\theta \leq 1$  for coinsurance)
- Behind the scene, computation is complex
  - Not a convex problem (like machine learning)
  - Requires simulation techniques to compute

### 3.3.2 An Upper Limit Contract

One risk, one retention parameter.

Consider an upper limit contract with fair costs, so the risk transfer cost is

$$RTC(u) = E(X) - E(X \wedge u) = \int_u^{\infty} (1 - F(z)) dz.$$

With this expression, we see that  $RTC(u)$  is monotonic in  $u$ . Thus  $RTC(u) \leq RTC_{max} \iff u \geq u_{max}$ . Using the value at risk for the risk measure, our problem is

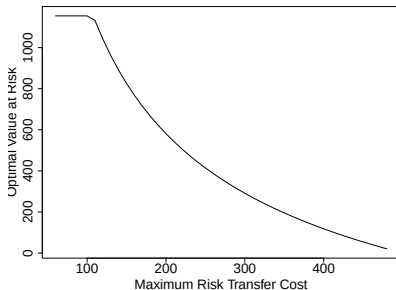
$$\begin{array}{ll} \text{minimize}_u & \min(F_{\alpha}^{-1}, u) \\ \text{subject to} & u \geq u_{max}. \end{array}$$

The risk measure is also monotonic in  $u$ , so you can check that

$$VaR_{\alpha}(X \wedge u)_{opt} = VaR_{\alpha}(X \wedge u_{max}) = \begin{cases} u_{max} & \text{if } u_{max} \leq F_{\alpha}^{-1} \\ F_{\alpha}^{-1} & \text{if } u_{max} > F_{\alpha}^{-1} \end{cases}.$$

## Example. Pareto Distribution.

- The quantile is  $F_{0.90}^{-1} = 1154.435$ .
- First use  $RTC_{max}$  to determine  $u_{max}$  (not shown)
- Then determine  $VaR_{\alpha}(X \wedge u)_{opt}$  (shown)
- Note that the value at risk is bounded by  $F_{0.90}^{-1}$ .



### 3.3.3 The Lagrangian

- The Lagrange method - minimize (or maximize) a function subject to equality constraints.
- Consider only those values of  $\theta$  so that  $RTC(\theta^*) = RTC_{max}$ .
- Then, the corresponding Lagrangian is

$$LA(\theta, LM) = RM[g(X; \theta)] + LM \{RTC(\theta) - RTC_{max}\}.$$

- One determines the parameters in  $\theta$  and the new parameter  $LM$  that optimize the function  $LA$ .
  - Turns a constrained problem into an unconstrained problem.
- To interpret the new parameter  $LM$ , take derivatives.

$$LM = -\frac{\partial_{\theta_i} RM[g(X; \theta)]}{\partial_{\theta_i} RTC(\theta)} = RM_i^2.$$

- This is a ratio of the differential changes (Section 2.3).
  - Called the **risk measure relative marginal** change, or  $RM^2$  change, for short.

### 3.3.4 A General Constrained Optimization Problem

$$\begin{array}{ll} \text{minimize}_{\theta} & f_0(\theta), \quad \theta \in R^p \\ \text{subject to} & f_{in,j}(\theta) \leq 0 \quad j = 1, \dots, m_{in} \\ & f_{eq,j}(\theta) = 0 \quad j = 1, \dots, m_{eq}. \end{array}$$

- In many instances, we are able to solve a constrained optimization problem numerically by relying on software.
- Extend the Lagrangian to include inequality constraints

$$LA(\theta, \mathbf{LMI}, \mathbf{LME}) = f_0(\theta) + \sum_{j=1}^{m_{in}} LMI_j f_{in,j}(\theta) + \sum_{j=1}^{m_{eq}} LME_j f_{eq,j}(\theta),$$

- $\mathbf{LMI} = (LMI_1, \dots, LMI_{m_{in}})'$  - Lagrange multiplier vector for inequalities
- $\mathbf{LME} = (LME_1, \dots, LME_{m_{eq}})'$  - Lagrange multiplier vector for equalities.

## Code for Several Optimization Runs - A Frontier!

```
meanSum = mean( g0_XL( u1=Inf, u2 = Inf) )
# Objective function to minimize expected shortfall
eval_f <- function(par, ...) {
  u1 <- par[1]
  u2 <- par[2]
  SumStats.fct( g0_XL( u1, u2 ) )[4]
}
# Inequality constraints (hin >= 0)
eval_g <- function(par, ...) {
  u1 <- par[1]
  u2 <- par[2]
  # Constraint: RTCmax - (meanSum_retained) >= 0
  RTCmax - (meanSum - SumStats.fct( g0_XL( u1, u2 ) )[1])
}
# Initial values for u.lim_alabama
u_init <- c(2, 1)

SumResults <- matrix(0, 9, 7)
# Loop through different RTCmax values
for (i_val in (1:9)) {
  RTCmax = (i_val/10) * meanSum
  # Run the optimization
  result_alabama <- auglag(
    par = u_init,
    fn = eval_f,
    hin = eval_g,
    lower = c(0, 0), # Add non-negativity constraint for u1 and u2
    control.outer=list(method="nlnmb",trace=FALSE) )
  # Extract the optimal portfolio proportions
  u.lim_alabama <- result_alabama$par
  SumResults[i_val,] <- c(RTCmax, SumStats.fct( g0_XL(u.lim_alabama[1],u.lim_alabama[2] ) ), u.lim_alabama
}
```

## Summary of Presentation

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- Risk Retention Functions
- Dependent Risks
- Constrained Optimization